Table 6.2 Comparison of other AI control systems

Control name	Description	Shortcomings	Control agentic AI?
Finite state machine [a] (FSM)	FSMs model AI using a set of states and transitions triggered by events or conditions.	FSMs can become unwieldy with increasing complexity.	FSMs aren't practical for agents because they don't scale well.
Decision tree [b]	Decision trees use a tree-like model of decisions and their possible consequences.	Decision trees can suffer from overfitting and lack generalization in complex scenarios.	Decision trees can be adapted and enhanced with behavior trees.
Utility-based system [b]	Utility functions evaluate and select the best action based on the current situation.	These systems require careful design of utility functions to balance priorities.	This pattern can be adopted within a behavior tree.
Rule-based system [a]	This set of if-then rules define the behavior of the AI.	These systems can become cumbersome with many rules, leading to potential conflicts.	These aren't very practical when paired with agentic systems powered by LLMs.
Planning system [c]	Planning systems generate a sequence of actions to achieve a specific goal using planning algorithms.	These systems are computationally expensive and require significant domain knowledge.	Agents can already implement such patterns on their own as we'll see in later chapters.
Behavioral cloning [c]	Behavioral cloning refers to learning policies by mimicking expert demonstrations.	This system may struggle with generalization to unseen situations.	This can be incorporated into behavior trees or into a specific task.
Hierarchical Task Network (HTN) [d]	HTNs decompose tasks into smaller, manageable subtasks arranged in a hierarchy.	These are complex to manage and design for very large tasks.	HTNs allow for better organization and execution of complex tasks. This pattern can be used for larger agentic systems.
Blackboard system [b]	These systems feature collaborative problem-solving using a shared blackboard for different subsystems.	These systems are difficult to implement and manage communication between subsystems.	Agentic systems can implement similar patterns using conversation or group chats/threads.
Genetic algorithm (GA) [d]	These optimization techniques are inspired by natural selection to evolve solutions to solve problems.	GAs are computationally intensive and may not always find the optimal solution.	GAs have potential and could even be used to optimize behavior trees.